

1. Administrative & Study Identification

Official Title: A Phase 1/2, Safety Confirmation, Placebo-controlled, Randomized Study of Nivolumab in Combination With Relatlimab and Bevacizumab in Treatment-naïve Advanced/Metastatic Hepatocellular Carcinoma **Study Title:** A Study of Nivolumab and Relatlimab in Combination With Bevacizumab in Advanced Liver Cancer **Short Title/Acronym:** RELATIVITY-106 **Protocol Number:** CA224-106 **Posting ID:** 218484283012645371 **NCT Number:** NCT05337137 **Sponsor / Company Name:** Bristol Myers Squibb **Investigational Product:** Nivolumab, Relatlimab, Bevacizumab (Trade names: Avastin, Vegzelma, Alymsys, Zirabev, Mvasi | ATC Codes: S01LA08, L01FG01) **Phase of Development:** PHASE1, PHASE2 **Trial Status:** ACTIVE_NOT_RECRUITING (Active but not currently recruiting) **Medical Monitor:** Bristol Myers Squibb Medical Team

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To evaluate the safety and effectiveness of triplet therapy of nivolumab, relatlimab, and bevacizumab versus nivolumab and bevacizumab in participants with untreated advanced/metastatic hepatocellular carcinoma (HCC). **Secondary Objectives:** To assess the efficacy of the combination therapies, primarily measured by Progression-Free Survival (PFS) and Overall Response Rate (ORR), particularly observing outcomes based on lymphocyte activation gene 3 (LAG-3) status.

2.2 Background & Rationale Advanced hepatocellular carcinoma has a poor prognosis and requires effective systemic therapies. While the combination of nivolumab and bevacizumab has shown efficacy in treating advanced liver cancer, this study investigates whether the addition of relatlimab (an anti-LAG-3 immune checkpoint inhibitor) can further enhance the immune response against tumor cells. The rationale is to establish whether this triple-drug combination can improve objective response rates and progression-free survival compared to the standard dual-drug combination in treatment-naïve patients.

3. Study Design & Methodology

3.1 Design Framework Study Type: Interventional **Control Type:** Placebo-controlled, Active Comparator **Blinding:** Double-blind **Randomization:** Randomized

3.2 Planned Enrollment & Duration Targeted Enrollment: 162 Number of Sites: Multiple global locations **Estimated Study Start:** April 2022 **Study Status:** Active, not recruiting

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. Histologically confirmed advanced/metastatic hepatocellular carcinoma (HCC). 2. Naïve to systemic therapy for advanced/metastatic HCC (prior neo-adjuvant or adjuvant immunotherapy is permitted if recurrence occurs ≥ 6 months after treatment completion and the case is discussed with BMS medical team). 3. Child-Pugh score of 5 or 6 (i.e., Child-Pugh A). 4. Eastern Cooperative Oncology Group (ECOG) performance status 0 or 1.

4.2 Exclusion Criteria 1. Known fibrolamellar HCC, sarcomatoid HCC, or mixed cholangiocarcinoma and HCC. 2. Prior allogenic stem cell or solid organ transplantation. 3. Untreated symptomatic central nervous system (CNS) metastases. 4. Clinically significant ascites as defined by: i) Prior ascites that required treatment and requires on-going prophylaxis, or ii) Current ascites requiring treatment.

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: Protocol-specified dosages on specified days per treatment cycle. * **Arm A:** Nivolumab + Relatlimab + Bevacizumab * **Arm B:** Nivolumab + Placebo + Bevacizumab **Route of Administration:** Intravenous (IV) **Duration of Treatment:** Cycles continue until disease progression, unacceptable toxicity, or withdrawal of consent.

5.2 Concomitant & Prohibited Therapy Permitted: Standard supportive care for symptoms. **Prohibited:** Concurrent systemic anti-cancer therapies, live vaccines, or other investigational agents not specified in the protocol.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures
1	2-4 weeks prior	Screening Informed Consent, Medical History, Tumor Biopsy/Imaging, Labs
2	Day 1 of each cycle	Treatment Cycles IV Administration of IP, Safety Labs, Vital Signs, AE Monitoring
3	Every 8-12 weeks	Efficacy Assessments Tumor imaging (CT/MRI) per RECIST v1.1
4	+ 30 days	End of Treatment Final Vital Signs, Exit Interview, IP Accountability, Safety Labs
5	Q12W	Follow-up Survival follow-up and tracking of subsequent therapies

7. Outcome Measures (Endpoints)

7.1 Primary Outcomes 1. Incidence of dose-limiting toxicities (DLTs). 2. Overall Response Rate (ORR) by Blinded Independent Central Review (BICR) per Response Evaluation Criteria in Solid Tumors (RECIST) v1.1.

7.2 Secondary Outcomes 1. Progression Free Survival (PFS) by BICR per RECIST v1.1 in all randomized participants that are lymphocyte activation gene 3 (LAG-3) positive. 2. PFS by BICR per RECIST v1.1 in all randomized participants. 3. ORR by BICR per RECIST v1.1 in all randomized participants that are LAG-3 positive.

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous collection from the signing of informed consent through 100 days post-last dose. Tracked for immune-mediated AEs (imAEs). **Serious Adverse Events (SAEs):** Must be reported to the sponsor within 24 hours of investigator awareness. **Data Monitoring Committee (DMC):** An independent Data Monitoring Committee (iDMC) assesses safety signals during the Phase 1 dose-confirmation and Phase 2 efficacy expansion stages.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: Intent-to-Treat (ITT) population for all efficacy outcomes; Safety Set for all participants who receive at least one dose of the investigational product. **Sample Size Justification:** Powered to detect a statistically significant improvement in PFS/ORR for the triplet arm (Arm A) compared to the doublet arm (Arm B). Targeted enrollment is 162 participants. **Handling of Missing Data:** Participants without a documented event or progression at the time of analysis will be censored at the date of their last evaluable tumor assessment.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and the Declaration of Helsinki. **Monitoring Plan:** Risk-based onsite and remote monitoring of clinical sites to ensure protocol adherence and data integrity. **Audit Trail:** Robust electronic Data Capture (eDC) system with full audit trails, eCRF locking mechanisms, and automated data clarification forms (DCFs).

11. Study Identifiers & Governance

Primary ID: {{CA224-106}} **Posting ID:** {{218484283012645371}}
Registry Number: {{NCT05337137}} **Sponsor/Lead Organization:** {{Bristol Myers Squibb}} **Study Title:** {{A Study of Nivolumab and Relatlimab in Combination With Bevacizumab in Advanced Liver Cancer}} **Official Regulatory Title:** {{A Phase 1/2, Safety Confirmation, Placebo-controlled, Randomized Study of Nivolumab in Combination With Relatlimab and Bevacizumab in Treatment-naïve Advanced/Metastatic Hepatocellular Carcinoma}}

12. Clinical Framework (Oncology Specific)

Disease Areas: {{Carcinoma, Hepatocellular}} **Preferred UMLS Name:** {{Hepatocellular Carcinoma}} **Semantic Type:** {{Neoplastic Process}} **Ontology Codes:** {{MeSH: D002277 | HPO: HP:0030731 | SNOMED: 1187425009}} **Condition Definition:** {{A malignant tumor arising from epithelial cells. Carcinomas that arise from glandular epithelium are called adenocarcinomas, those that arise from squamous epithelium are called squamous cell carcinomas, and those that arise from transitional epithelium are called transitional cell carcinomas.}} **Primary Condition:** {{Advanced/Metastatic Hepatocellular Carcinoma (HCC)}} **Diagnosis Threshold:** {{Histologically confirmed HCC; Child-Pugh A; Systemic therapy-naïve}} **Medical Methodology:** {{Triplet Immune Checkpoint & Anti-angiogenic Inhibition}} **Optimization Target:** {{Overall Response Rate (ORR) and Progression-Free Survival (PFS)}}

13. Study Procedures & Methodology

Management Pathway: {{Standard oncology clinical pathway for advanced HCC}} **Education Protocol (HCP):** {{Investigator training on triplet regimen safety, immune-mediated AE management, and infusion protocols}} **Education Protocol (Patient):** {{Informed consent process, treatment schedule adherence, and early AE reporting education}} **Quality Audit Mechanism:** {{Blinded Independent Central Review (BICR) for imaging & centralized safety monitoring}}

14. Observation & Follow-Up Schedule

Total Duration: {{Treatment until progression/toxicity + subsequent Survival Follow-up}} **Onsite Visit Cadence:** {{Day 1 of every treatment cycle (e.g., Q3W/Q4W)}} **Remote Monitoring Frequency:** {{Between cycles as clinically indicated}} **Checklist Review Interval:** {{Tumor imaging assessments every 8 to 12 weeks}}

15. Sign-off & Approval

	Role		Name		Signature		Date			:---		:---		:---		:---	
	Principal Investigator		[Insert Name]		_____		[DD-MMM-YYYY]										
	Clinical Operations Lead		[Insert Name]		_____		[DD-MMM-YYYY]										
		Lead Statistician		[Insert Name]		_____		[DD-MMM-YYYY]									